A 3.00g sample containing Fe_3O_4, Fe_2O_3 and an inert impure substance, is treated with excess of KI solution in presence of dilute H_2SO_4. The entire iron is converted into FE^{2+} along with the liberation of iodine. The resulting solution is diluted to 100 ml. A 20 ml of the diluted solution require 11.0 ml of 0.5 M $Na_2S_2O_3$ solution to reduce the iodine present. A 50 ml of the diluted solution, after complete extraction of the iodine requires 12.80 ml of 0.25 M $KMnO_4$ solution in dilute H_2SO_4 medium for the oxidation of Fe^{2+}. Calculate the percentages of Fe_2O_3 and Fe_3O_4 in the original sample.

Sol^n.

$\overset{+8/3}{Fe_3O_4} + \overset{+3}{Fe_2O_3} + \text{Impurities}$ 3 g

x mol, y mol, vf = 2

$vf = \frac{2}{3} \times 3 = 2$

$KI + H_2SO_4(aq)$

$Fe^{2+}(aq) + I_2$ 20 ml

vf = 1, $(3x+2y)$ 100ml

$Na_2S_2O_3 \rightarrow I^- + S_4O_6^{2-}$ vf = 1

11ml, 0.5M

50 ml

12.8 ml, 0.25M $KMnO_4 + H^+ \rightarrow Mn^{2+} + Fe^{3+}$

$x \times 2 + y \times 2 = \left(\frac{11}{1000} \times 0.5 \times 1\right) \times \frac{100}{5}$ —— ①

$(3x+2y) \times 1 = \left(\frac{12.8}{1000} \times 0.25 \times 5\right) \times \frac{100}{50}$ —— ②

$x = \checkmark$

$y = \checkmark$

$\% Fe_3O_4 = \frac{x \times (3 \times 56 + 64)}{3} \times 100$

$= 34.8\%$ Ans

$\% Fe_2O_3 = \frac{y \times (2 \times 56 + 48)}{3} \times 100$

49.3%

A sample of hard water contains 96 ppm of SO_4^{2-} and 183 ppm of HCO_3^-, with Ca^{2+} as the only cation. How many moles of CaO will be required to remove HCO_3^- from 1000 kg of this water?

→ 1.5 mol Ans → 10^6 g

If 1000 kg of this water is treated with the amount of CaO calculate aboce, what will be the conentration (in ppm) of residual Ca^{2+} ions (Assume $CaCO_3$ to be completely insoluble in water)?

→ 40 ppm Ans

If the Ca^{2+} ions in one litre of the treated water are completely exchanged with hydrogen ions, what will be its pH (one ppm means one part of the substance in one million part of water, weight/weights)?

→ ≈ 2.7 Ans.

Solⁿ: SO_4^{2-} : 96 ppm, $CaSO_4(aq)$ → 1 mol ← causes permanent hardness

HCO_3^- : 183 ppm, $Ca(HCO_3)_2(aq)$ → 1.5 mol ← temporary hardness

$\frac{96}{96} = 1$ mol $\frac{183}{61} = 3$ mol

$Ca(HCO_3)_{2}(aq) + CaO(s) \rightarrow 2CaCO_3\downarrow + H_2O(l)$
1.5 mol 1.5 mol Ans

(ii) Ca^{2+} → 1 mol in 10^6 g of water → 40 ppm Ans

$d_{H_2O(l)} = 1$ g/mL.

1000 kg water $\equiv$ 1000 lit.

$[Ca^{2+}] = 10^{-3}$ mol/lit

$[H^+] = 2\times10^{-3}$ M

pH = 2.7 Ans

Calcium phosphide (Ca_3P_2) formed by reacting calcium orthophosphate [$Ca_3(PO_4)_2$] with magnesium was hydrolysed by water. The evolved phosphine (PH_3) was burnt in air to yield phosphorus pentoxide (P_2O_5). (At. Wt. Mg = 24, P = 31)

$$Ca_3(PO_4)_2 + 8Mg \longrightarrow Ca_3P_2 + 8MgO \quad ......(1)$$

$$Ca_3P_2 + 6H_2O \longrightarrow 3Ca(OH)_2 + 2PH_3 \quad ......(2)$$

$$2PH_3 + 4O_2 \longrightarrow P_2O_5 + 3H_2O \quad ......(3)$$

$$MgO + P_2O_5 \longrightarrow Mg(PO_3)_2 \quad ......(4)$$

Magnesium metaphosphate

Now balanced eq^n.

Calculate moles of $Mg(PO_3)_2$ produced if

(i) 80 moles of Mg and 20 moles of O_2 are used with sufficient amount of other reactants

Ans: 5 mol

(ii) 80 moles of Mg and 20 moles of O_2 are used and reaction (2) has yield 50% → 5 mol Ans.

(ii) $Ca_3(PO_4)_2 + 8Mg$ (80 mol) $\longrightarrow Ca_3P_2 + 8MgO$ (10 mol)

$Ca_3P_2 + 6H_2O \longrightarrow 3Ca(OH)_2 + 2PH_3$
10 mol; 50% → 10 mol

$2PH_3 + 4O_2 \longrightarrow P_2O_5 + 3H_2O$
10 mol, 20 mol, 5 mol

$MgO + P_2O_5 \longrightarrow Mg(PO_3)_2$
5 mol

5 mol

Sol^n: (i) Ca_3P_2 = 10 mol.
PH_3 = 20 mol.

$2PH_3 + 4O_2 \longrightarrow P_2O_5 + 3H_2O$
20 20 (L.R.) → 5 mol

$Mg(PO_3)_2$ → 5 mol

Ans

When 10 ml of acetic acid (density = 0.8 g/ml) is mixed with 40 ml of water (density = 1 g/ml) at a certain temperature, the final solution is found to have a density of $\frac{96}{98}$ g/ml. The percent change in total volume on mixing is → Ans : 2 %

Sol^n:

$CH_3COOH(l)$: 10 ml → $10 \times 0.8 = 8\,g$

$H_2O(l)$: 40 ml → $40 \times 1 = 40\,g$

$\underbrace{CH_3COOH + H_2O}$

↓ after mixing

$d_{solution} = \frac{96}{98}\,g/ml = \frac{(8+40)}{V_f} \Rightarrow V_f = \frac{\cancel{48}}{\cancel{96}_2} \times 98 = 49\,ml$

Expected volume after mixing $= 10 + 40 = 50\,ml$

Actual final vol $= 49\,ml$

% change in vol $= \frac{(50-49)}{50} \times 100 = 2\,\%$ Ans.

Household bleach contains hypochlorite ion, which is formed when chlorine dissolves in water. To determine the concentration of hypochlorite in the bleach, the solution is first treated with a KI solution. The iodine liberated can be determined by titration with a standard thiosulphate solution. A 25 ml of certain household bleach requires 17.4 ml of a 0.02 M-$Na_2S_2O_3$ solution for titration. The mass of chlorine dissolved in one litre of the bleach solution is

(A) 0.1392 g (B) 0.494 g Ans (C) 9.88 g (D) 0.278 g

$Cl_{2(g)} + H_2O(l) \rightarrow HCl + HOCl$ (O.A) hypochlorous acid

$F_2 > Cl_2 > Br_2 > I_2$ decreasing oxidising power.

Sol^n:

$\overset{0}{Cl_2}(aq) \xrightarrow{KI(aq)\ (\text{excess})} I_2 + \overset{-1}{Cl^-}(aq)$ intermediate

Cl_2: 25 ml, $v_f = 2$

$I_2 \xrightarrow{\text{Standard hypo sol}^n\ (\overset{+2}{Na_2S_2O_3})} NaI + Na_2\overset{+2.5}{S_4O_6}$

$Na_2S_2O_3$: $v_f = 1$, 17.4 ml, 0.02 M

Eq. of Cl_2 = Eq. of I_2 = Eq. of hypo.

$\frac{W_{Cl_2}}{71} \times 2 = \frac{17.4}{1000} \times 0.02 \times 1$

$W_{Cl_2} = \frac{17.4 \times 0.01}{1000} \times 71$ g present in 25 ml bleach,

$(W_{Cl_2})_{\text{in 1 lit}} = 17.4 \times 10^{-5} \times 71 \times \frac{1000}{25}$

$= \frac{17.4 \times 0.01 \times 71}{25}$

$= 0.494$ g Ans.

→ Primary titrant : $K_2Cr_2O_7$, $\underbrace{H_2C_2O_4,\ \text{Mohr salt}}$, $Na_2S_2O_3 \cdot 5H_2O$

used to standardize $KMnO_4$ sol^n ← secondary titrant.

Secondary titrant : $KMnO_4$, $Na_2Cr_2O_7$,

→ $\overset{+7}{KMnO_4}$

- $\xrightarrow{5}$ Acidic (H_2SO_4) → $\overset{+2}{Mn^{2+}}$, I_2 etc.
- $\xrightarrow{3}$ neutral/basic → $\overset{+4}{MnO_2}\downarrow$
- $\xrightarrow{1}$ strongly basic → $\overset{+6}{MnO_4^{2-}}$

$v_f = 1$

* HCl should not be used for making $KMnO_4$ medium acidic

$MnO_4^- + HCl \rightarrow Cl_2\uparrow + Mn^{2+}$